

Track A — Genomic Baseline with 6-Channel Walsh–Hadamard Transform

Prior Art Defensive Publication and Technical Report

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Abstract

This document specifies *Track A — Genomic Baseline*, a 2–3 week experimental protocol that applies a six-channel Walsh–Hadamard Transform (WHT) spectral decomposition to a publicly available codon-usage dataset for *Escherichia coli* K-12 (approximately 4,000 coding sequences). For each gene, a pairwise spectral distance Δh is computed between native codon usage and a GC3-controlled reference ensemble. A logistic classifier is then trained on Δh and contrasted with a classifier trained on GC3-only features, and the absolute accuracy delta is measured. This constitutes a Go/No-Go gate for the Track A hypothesis that WHT-based multiplicity structure captures biologically meaningful codon-usage signal beyond GC3 composition alone. The report is written as a defensive prior-art publication to prevent third-party patent claims on the described methodology.

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1 Executive Summary

Track A is designed as the fastest executable testbed for Multiplicity Theory in a concrete biological domain: codon usage bias in *E. coli* K-12.¹ The central question is whether a six-channel Walsh–Hadamard spectral decomposition of codon usage vectors, followed by an appropriately defined Δh comparison to GC3-matched null models, yields predictive structure not already explained by GC3 composition alone.

The core experimental steps are:

1. Download a curated CDS set for *E. coli* K-12 (K-12 MG1655) with roughly 4,000 protein-coding genes and clean the sequences (frame-correct, multiple-of-three lengths, standard genetic code).
2. For each gene, compute:
 - a codon-usage vector $c_g \in \mathbb{R}^{61}$ (excluding stops),
 - GC content overall and at third codon positions (GC3_g), and optionally other base-composition metrics.[web:303][web:304]
3. Apply a six-channel WHT projection mapping c_g into six interaction channels associated with multiplicity-theoretic axes (e.g., synonymous vs. non-synonymous, purine/pyrimidine partitions, etc.).
4. For each gene, generate a GC3-controlled reference ensemble via codon reshuffling that preserves amino-acid sequence and GC3 profile; compute the corresponding WHT spectrum and define a per-gene Δh_g as the normed distance between native and reference spectra.
5. Construct two logistic regression classifiers for a ground-truth binary label (e.g., native vs. foreign codon-usage status, or high vs. low expression class): one using only GC3-based features, and one using GC3 plus Δh_g (or Δh -derived features).
6. Quantify the absolute accuracy delta, $\Delta \text{Acc} = \text{Acc}(\text{GC3} + \Delta h) - \text{Acc}(\text{GC3-only})$, using cross-validation.

A positive ΔAcc supports the Track A thesis that Δh captures non-trivial, GC3-independent structure in codon usage; a null or negligible ΔAcc is a “No-Go” outcome that falsifies the specific spectral hypothesis at this resolution.

This document defines the mathematical framework, concrete pipeline, and representative code patterns with sufficient detail to re-implement the experiment and to establish defensive prior art.

2 Data Model and Preprocessing

2.1 Genome and CDS Selection

Track A assumes the use of a standard, publicly available CDS annotation for *E. coli* K-12 (e.g., MG1655) from NCBI or a similar source. Multiple tools and datasets exist for codon usage analysis in *E. coli*, including utilities such as GCUA and coRdon that compute RSCU, GC3, and other metrics over CDS sets.[web:303][web:304][web:306]

¹Codon usage and GC3-based analyses for *E. coli* K-12 are standard and widely used in comparative genomics; see, e.g., general codon usage tools such as GCUA and related GC3-focused pipelines, as well as public codon tables for this organism.[web:303][web:317]

Requirements.

- All CDS must be in-frame, length divisible by three, and annotated using the standard bacterial genetic code.
- Start and stop codons may be handled separately; in this specification stops are excluded from codon-usage vectors (yielding a 61-dimensional space).
- Genes shorter than a minimal length threshold (e.g., fewer than 100 codons) may be filtered out for stability.

Let $\mathcal{G}$ denote the set of genes retained after quality control, with $|\mathcal{G}| = G \approx 4,000$.

2.2 Codon-Usage Vectors

Let $\mathcal{C}$ be the set of 61 sense codons. For each gene $g \in \mathcal{G}$, let $\text{count}_g(c)$ be the count of codon $c \in \mathcal{C}$ in that gene.

Definition 2.1 (Raw codon-usage vector). *The raw codon-usage vector for gene g is*

$$u_g(c) = \text{count}_g(c), \quad c \in \mathcal{C},$$

which can be arranged as $u_g \in \mathbb{R}^{61}$.

Definition 2.2 (Normalized codon-usage vector). *Let $L_g = \sum_{c \in \mathcal{C}} u_g(c)$ be the codon length of gene g . The normalized codon-usage vector is*

$$c_g(c) = \frac{u_g(c)}{L_g}, \quad c \in \mathcal{C},$$

yielding $c_g \in \Delta^{60}$, the 61-dimensional probability simplex.

2.3 GC3 and Other GC-Based Metrics

For each gene g , let GC_g be the total GC fraction and GC3_g the GC fraction at third codon positions.

Definition 2.3 (GC3 content). *Let $N_{3,\text{G/C}}(g)$ be the number of positions at third codon sites that are G or C, and let $N_{3,\text{total}}(g)$ be the total number of third codon positions (i.e., L_g). Then*

$$\text{GC3}_g = \frac{N_{3,\text{G/C}}(g)}{N_{3,\text{total}}(g)}.$$

GC3 and related metrics are standard in codon usage bias analysis, and multiple existing packages compute them directly from codon counts.[web:303][web:304][web:308]

Additional features may include overall GC, GC1, GC2, effective number of codons, or CAI, but the minimal Track A baseline focuses on GC3 (possibly plus overall GC as a secondary feature).

3 Six-Channel WHT Representation

3.1 Walsh–Hadamard Transform Basics

The Walsh–Hadamard Transform (WHT) provides an orthogonal decomposition over a hypercube, often used in epistasis and combinatorial genetics to decompose additive vs. interaction effects across binary factors. In Track A, WHT is repurposed as a spectral transform over codon features, with six channels representing specific multiplicity-theoretic axes.

Formally, the standard WHT on 2^n inputs can be written as multiplication by the $2^n \times 2^n$ Hadamard matrix $H^{(n)}$ with entries ± 1 . For our purposes, codons are grouped and mapped into six aggregated channels, rather than applying a full 2^{61} -dimensional transform.

3.2 Channel Construction

Let there be a fixed partition of $\mathcal{C}$ into six groups $\mathcal{C}_1, \dots, \mathcal{C}_6$. The grouping is chosen to reflect biologically meaningful dimensions such as:

- GC3 vs. AT3 at synonymous sites,
- amino-acid hydrophobicity classes,
- purine vs. pyrimidine at first or second positions,
- or other structured partitions informed by multiplicity labels.

The exact grouping is part of the Track A design and must be fixed before analysis.

Definition 3.1 (Channel aggregation). *Given a normalized codon-usage vector c_g , define the aggregated channel intensities*

$$a_{g,i} = \sum_{c \in \mathcal{C}_i} c_g(c), \quad i = 1, \dots, 6.$$

Let $\mathbf{a}_g = (a_{g,1}, \dots, a_{g,6})^\top \in \mathbb{R}^6$.

Definition 3.2 (6-channel Walsh–Hadamard transform). *Let H_6 be a fixed 6×6 orthogonal matrix with entries $\pm 1/\sqrt{6}$, representing a WHT-like transform on six channels (e.g., the first six rows of a normalized Hadamard matrix extended or adapted to a non-power-of-two dimension). The 6-channel WHT spectrum for gene g is*

$$h_g = H_6 \mathbf{a}_g \in \mathbb{R}^6.$$

Remark 3.3. *The Track A protocol is agnostic to the exact choice of H_6 so long as it is orthogonal, has entries of comparable magnitude, and is fixed across all genes. The key idea is to obtain a small set of orthogonal coordinates capturing combinatorial structure across the six aggregated codon channels.*

4 GC3-Controlled Null Ensemble and Δh

4.1 GC3-Controlled Randomization

To disentangle GC3-driven effects from higher-order codon structure, Track A constructs a null model of codon usage per gene that preserves:

- the amino-acid sequence,
- the gene length,
- the GC3 content (up to small binning or tolerance),

while randomizing codon choices within constraints.

Definition 4.1 (GC3-preserving codon reshuffling). *For gene g with amino-acid sequence $\alpha_1, \dots, \alpha_{L_g}$ and GC3 content GC3_g , define a procedure that samples random synonymous codons at each position such that:*

- *the chosen codon encodes the correct amino acid α_i ,*
- *the overall GC3 count matches a target consistent with GC3_g (exact match or small tolerance).*

Repeating this sampling R times produces an ensemble of reshuffled codon-usage vectors $\{c_{g,r}^{\text{null}}\}_{r=1}^R$.

Tools such as `codonuse` illustrate closely related ideas, where null codon usage is generated by randomizing codons while controlling GC3.[web:305]

4.2 Null WHT Spectrum and Distance

For each gene and each null replicate, compute the 6-channel spectrum as in the native case.

Definition 4.2 (Null WHT spectra). *For gene g and null replicate r , define*

$$h_{g,r}^{\text{null}} = H_6 \mathbf{a}_{g,r}^{\text{null}} \in \mathbb{R}^6,$$

where $\mathbf{a}_{g,r}^{\text{null}}$ is computed from $c_{g,r}^{\text{null}}$ by channel aggregation.

Definition 4.3 (Mean null spectrum and Δh). *Define the mean null spectrum*

$$\bar{h}_g^{\text{null}} = \frac{1}{R} \sum_{r=1}^R h_{g,r}^{\text{null}}.$$

Then define

$$\Delta h_g = \left\| h_g - \bar{h}_g^{\text{null}} \right\|_2,$$

or more generally, the vector of component-wise deviations

$$\Delta \mathbf{h}_g = h_g - \bar{h}_g^{\text{null}} \in \mathbb{R}^6.$$

Remark 4.4. *The scalar Δh_g is a single summary statistic, while $\Delta \mathbf{h}_g$ provides six-dimensional features. Both can be used as inputs to downstream classifiers.*

5 Classification Setup and Go/No-Go Gate

5.1 Ground-Truth Labels

Track A requires a binary label per gene to serve as ground truth for supervised classification. Possibilities include:

- Native vs. foreign codon usage, as determined by an existing axis-of-variation analysis over *E. coli* K-12 genes (e.g., classification into native vs. horizontally transferred genes).[web:321]
- High vs. low expression classes determined from external expression data (e.g., ribosomal vs. lowly expressed genes).

For definiteness, let $y_g \in \{0, 1\}$ be the binary label for gene g , where $y_g = 1$ might denote “native” or “highly expressed”.

5.2 Feature Sets

Define two feature sets per gene g :

GC3-only features.

$$X_g^{\text{GC3}} = (\text{GC3}_g) \in \mathbb{R},$$

or extended to a small GC-based vector such as $(\text{GC}_g, \text{GC3}_g) \in \mathbb{R}^2$.

GC3 + Δh features.

$$X_g^{\text{GC3}+\Delta h} = \begin{cases} (\text{GC3}_g, \Delta h_g) \in \mathbb{R}^2, & \text{scalar } \Delta h \text{ case,} \\ (\text{GC3}_g, \Delta h_{g,1}, \dots, \Delta h_{g,6}) \in \mathbb{R}^7, & \text{vector } \Delta \mathbf{h}_g \text{ case.} \end{cases}$$

5.3 Logistic Regression Models

Definition 5.1 (GC3-only logistic model). *Let $\theta^{\text{GC3}} \in \mathbb{R}^{d_1}$ with $d_1 = 1$ or 2 depending on the GC3 feature dimension. The GC3-only logistic model is*

$$\Pr(y_g = 1 \mid X_g^{\text{GC3}}) = \sigma((\theta^{\text{GC3}})^\top X_g^{\text{GC3}}), \quad \sigma(z) = \frac{1}{1 + e^{-z}}.$$

Definition 5.2 (GC3 + Δh logistic model). *Let $\theta^{\text{GC3}+\Delta h} \in \mathbb{R}^{d_2}$ with $d_2 = 2$ or 7. The expanded logistic model is*

$$\Pr(y_g = 1 \mid X_g^{\text{GC3}+\Delta h}) = \sigma((\theta^{\text{GC3}+\Delta h})^\top X_g^{\text{GC3}+\Delta h}).$$

These models can be trained via standard ℓ_2 -regularized maximum likelihood or stochastic gradient methods, as commonly used for sequence-based logistic classifiers.[web:307]

5.4 Accuracy and Absolute Delta

For each model, evaluate accuracy via cross-validation:

Definition 5.3 (Cross-validated accuracy). *Partition $\mathcal{G}$ into K folds. For each fold, train the model on $K - 1$ folds and evaluate on the held-out fold, then average. Denote the resulting accuracies as Acc_{GC3} and $\text{Acc}_{\text{GC3}+\Delta h}$.*

Definition 5.4 (Absolute accuracy delta). *The Track A Go/No-Go metric is*

$$\Delta \text{Acc} = \text{Acc}_{\text{GC3}+\Delta h} - \text{Acc}_{\text{GC3}}.$$

Remark 5.5 (Go/No-Go interpretation). *A substantially positive ΔAcc (beyond noise and confidence intervals) indicates that the WHT-derived Δh feature carries predictive information not captured by GC3 alone, supporting the Track A spectral hypothesis. A negligible or zero ΔAcc is a “No-Go” outcome, suggesting that codon-usage signal relevant to the chosen labels is well-approximated by GC3 composition and that the proposed 6-channel WHT does not add measurable discriminative power under this design.*

6 Representative Code Snippet (Python)

This section provides representative pseudocode in Python (using NumPy and scikit-learn) for the Track A pipeline. It is not meant to be a drop-in implementation but rather a concrete sketch of the key steps.

6.1 Feature Extraction and Δh

```
import numpy as np
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import StratifiedKFold
from Bio import SeqIO # from Biopython

CODONS = [ # 61 sense codons, stops excluded
    "AAA", "AAC", "AAG", "AAT", # ...
    # (fill with all codons except TAA, TAG, TGA)
]

CODON_IDX = {c: i for i, c in enumerate(CODONS)}

# Example 6x6 orthogonal transform (placeholder { must be fixed beforehand)
H6 = (1.0 / np.sqrt(6)) * np.array([
    [ 1,  1,  1,  1,  1,  1],
    [ 1, -1,  1, -1,  1, -1],
    [ 1,  1, -1, -1,  1,  1],
    [ 1, -1, -1,  1,  1, -1],
    [ 1,  1,  1, -1, -1, -1],
    [ 1, -1,  1,  1, -1,  1],
])

# User-defined mapping of codons to 6 channels
def codon_channel(codon: str) -> int:
    # TODO: define biologically meaningful partition
    # For now, a placeholder mapping based on GC3 etc.
    # Return an integer 0..5.
    raise NotImplementedError

def gc3_content(seq: str) -> float:
    thirds = seq[2::3]
    gc = sum(1 for b in thirds if b in "GCgc")
    return gc / max(len(thirds), 1)
```

```

def codon_usage_vector(seq: str) -> np.ndarray:
    counts = np.zeros(len(CODONS), dtype=float)
    for i in range(0, len(seq) - 2, 3):
        codon = str(seq[i:i+3]).upper()
        if codon in CODON_IDX:
            counts[CODON_IDX[codon]] += 1
    total = counts.sum()
    if total > 0:
        counts /= total
    return counts

def six_channel_projection(codon_vec: np.ndarray) -> np.ndarray:
    channel_sums = np.zeros(6, dtype=float)
    for codon, idx in CODON_IDX.items():
        ch = codon_channel(codon)
        channel_sums[ch] += codon_vec[idx]
    return H6 @ channel_sums # 6D spectrum

def random_gc3_preserving_sequence(seq: str, target_gc3: float) -> str:
    # Placeholder; real implementation must enforce amino acid and GC3 constraints.
    raise NotImplementedError

def compute_delta_h_for_gene(seq: str, R: int = 100) -> float:
    # Native spectrum
    cu_native = codon_usage_vector(seq)
    h_native = six_channel_projection(cu_native)

    # GC3-controlled null ensemble
    gc3 = gc3_content(seq)
    h_null = []
    for _ in range(R):
        seq_null = random_gc3_preserving_sequence(seq, gc3)
        cu_null = codon_usage_vector(seq_null)
        h_null.append(six_channel_projection(cu_null))
    h_null = np.vstack(h_null)
    h_null_mean = h_null.mean(axis=0)

    delta = np.linalg.norm(h_native - h_null_mean)
    return float(delta)

```

6.2 Logistic Regression and Accuracy Delta

```

def run_trackA_classification(cds_fasta_path, labels_dict, n_splits=5):
    seqs = []
    gc3_vals = []
    delta_h_vals = []
    ys = []

```

```

for rec in SeqIO.parse(cds_fasta_path, "fasta"):
    gene_id = rec.id
    if gene_id not in labels_dict:
        continue
    seq = str(rec.seq)
    if len(seq) % 3 != 0:
        continue # enforce in-frame

    gc3 = gc3_content(seq)
    d_h = compute_delta_h_for_gene(seq)

    seqs.append(seq)
    gc3_vals.append(gc3)
    delta_h_vals.append(d_h)
    ys.append(labels_dict[gene_id])

gc3_vals = np.array(gc3_vals).reshape(-1, 1)
delta_h_vals = np.array(delta_h_vals).reshape(-1, 1)
ys = np.array(ys)

X_gc3 = gc3_vals
X_gc3_dh = np.hstack([gc3_vals, delta_h_vals])

clf_gc3 = LogisticRegression(max_iter=1000)
clf_gc3_dh = LogisticRegression(max_iter=1000)

skf = StratifiedKFold(n_splits=n_splits, shuffle=True, random_state=42)

acc_gc3, acc_gc3_dh = [], []

for train_idx, test_idx in skf.split(X_gc3, ys):
    X_tr_gc3, X_te_gc3 = X_gc3[train_idx], X_gc3[test_idx]
    X_tr_dh, X_te_dh = X_gc3_dh[train_idx], X_gc3_dh[test_idx]
    y_tr, y_te = ys[train_idx], ys[test_idx]

    clf_gc3.fit(X_tr_gc3, y_tr)
    clf_gc3_dh.fit(X_tr_dh, y_tr)

    acc_gc3.append(clf_gc3.score(X_te_gc3, y_te))
    acc_gc3_dh.append(clf_gc3_dh.score(X_te_dh, y_te))

acc_gc3 = float(np.mean(acc_gc3))
acc_gc3_dh = float(np.mean(acc_gc3_dh))
delta_acc = acc_gc3_dh - acc_gc3

return acc_gc3, acc_gc3_dh, delta_acc

```

This code illustrates the minimal Track A computation of Δh per gene and the comparison of GC3-only vs. GC3+ Δh logistic models under cross-validation.

7 Implementation Timeline and Practical Notes

7.1 Timeline (2–3 Weeks)

A realistic 2–3 week schedule for Track A is:

- **Week 1:** Acquire and clean CDS for *E. coli* K-12, implement codon counting and GC3 computation, implement 6-channel mapping and WHT projection.
- **Week 2:** Implement GC3-preserving randomization, compute Δh per gene, assemble labels, and run initial logistic regression experiments.
- **Week 3 (optional):** Refine channel design, explore alternative labels (e.g., native vs. foreign axes), and perform robustness checks (different randomization schemes, different WHT bases, etc.).

7.2 Design Choices and Extensions

Key design choices that can be varied in follow-up experiments include:

- Alternative channel partitions $\mathcal{C}_i$ (e.g., focusing more tightly on GC3-synonymous groups).
- Using full $\Delta \mathbf{h}_g$ (6 channels) rather than scalar Δh_g .
- Incorporating additional GC-based features, RSCU, or CAI as baselines.
- Switching to other linear classifiers (e.g., SVMs or elastic-net logistic regression) while preserving the GC3 vs. GC3+ Δh comparison.

8 Defensive Publication Statement

Scope

This document constitutes a defensive publication covering the following Track A contributions:

1. The use of a six-channel WHT spectral decomposition over codon-usage vectors for genome-wide analysis of *E. coli* K-12 genes.
2. The definition of a GC3-preserving null ensemble per gene and the resulting per-gene Δh as the normed difference between native and null WHT spectra.
3. The use of Δh as a covariate in a logistic regression classifier alongside GC3 (and related GC-based features) to test for non-GC3 codon-usage signal.
4. The Go/No-Go gate defined in terms of the absolute accuracy delta between GC3-only and GC3+ Δh models in supervised classification of gene labels.

Intent

By publishing this specification with technical and mathematical detail sufficient for independent re-implementation, the authors intend to create prior art that prevents third parties from obtaining patent rights over implementations of Track A as described here, including variants that differ only in straightforward parameter tuning or channel selection.

center

A Hadamard Matrices and 6-Channel WHT Approximation

A.1 Classical Hadamard Matrices

The Walsh–Hadamard transform is defined by a family of *Hadamard matrices*, which are real orthogonal matrices with entries of equal magnitude.[?, ?]

Definition A.1 (Hadamard matrix). *A (Sylvester-type) Hadamard matrix of order $n = 2^m$ is a matrix $H_n \in \{-1, 1\}^{n \times n}$ defined recursively by*

$$H_1 = [1], \quad H_{2n} = \begin{bmatrix} H_n & H_n \\ H_n & -H_n \end{bmatrix}.$$

Its normalized version is

$$\hat{H}_n = \frac{1}{\sqrt{n}} H_n \in \mathbb{R}^{n \times n}.$$

Proposition A.2 (Orthogonality and norm preservation). *For each $n = 2^m$,*

$$\hat{H}_n \hat{H}_n^\top = I_n = \hat{H}_n^\top \hat{H}_n,$$

so $\hat{H}_n$ is orthonormal. In particular, for any $x \in \mathbb{R}^n$,

$$\left\| \hat{H}_n x \right\|_2 = \|x\|_2.$$

Proof. The orthogonality property follows by induction on n , using the block recursion and the fact that for the base case $H_1 = [1]$, $\hat{H}_1 = [1]$ is trivially orthonormal. Standard references on the Walsh–Hadamard transform derive these identities explicitly and highlight that the normalized matrix has entries of magnitude $n^{-1/2}$ and is orthonormal.[?, ?] The 2-norm preservation is a direct consequence of orthonormality. $\square$

A.2 Restriction to Six Channels

Track A does not operate in a power-of-two dimension; instead, it aggregates 61 codon frequencies into six biologically motivated channels and then applies a *6-channel WHT-like* transformation.

Definition A.3 (6-channel transform). *A matrix $H_6 \in \mathbb{R}^{6 \times 6}$ is called a 6-channel WHT approximation if:*

1. *H_6 is orthogonal: $H_6 H_6^\top = I_6 = H_6^\top H_6$.*
2. *Entries of H_6 have equal magnitude up to a common scaling (e.g., $\pm 1/\sqrt{6}$), mirroring Hadamard structure.*

Given a channel intensity vector $a_g \in \mathbb{R}^6$ for gene g , the 6-channel spectrum is

$$h_g = H_6 a_g \in \mathbb{R}^6.$$

Proposition A.4 (Norm preservation for 6-channel transform). *If H_6 is orthogonal, then for each gene g ,*

$$\|h_g\|_2 = \|a_g\|_2.$$

Proof. Orthogonality gives $H_6^\top H_6 = I_6$. For any a_g ,

$$\|h_g\|_2^2 = h_g^\top h_g = a_g^\top H_6^\top H_6 a_g = a_g^\top a_g = \|a_g\|_2^2.$$

Taking square roots yields the claim. $\square$

Remark A.5. *Although a true Hadamard matrix exists only for dimensions 2^m , any real orthogonal 6×6 matrix can serve as the spectral mixing matrix for Track A, and its orthogonality is all that is required for the norm bounds and distance properties used in the main text.[?, ?]*

B Codon-Usage Vectors and Channel Aggregation

B.1 Codon-Usage Geometry

Let $\mathcal{C}$ be the set of 61 sense codons. For each gene g , the normalized codon-usage vector $c_g \in \mathbb{R}^{61}$ satisfies

$$c_g(c) \geq 0 \quad \forall c \in \mathcal{C}, \quad \sum_{c \in \mathcal{C}} c_g(c) = 1,$$

so c_g lies in the 61-dimensional simplex.

We partition $\mathcal{C}$ into six disjoint channel sets $\mathcal{C}_1, \dots, \mathcal{C}_6$, chosen to reflect biologically meaningful groupings (GC3, amino-acid classes, etc.) as described in the main specification.

Definition B.1 (Channel aggregation map). *The channel aggregation map $A : \mathbb{R}^{61} \rightarrow \mathbb{R}^6$ is defined by*

$$a_{g,i} = \sum_{c \in \mathcal{C}_i} c_g(c), \quad i = 1, \dots, 6,$$

and $a_g = (a_{g,1}, \dots, a_{g,6})^\top$.

Proposition B.2 (Norm bounds under aggregation). *Let $\|\cdot\|_2$ be the Euclidean norm on both $\mathbb{R}^{61}$ and $\mathbb{R}^6$. Then for each gene g ,*

$$\|a_g\|_2 \leq \|c_g\|_2 \leq 1.$$

Proof. Denote the aggregation matrix by $A \in \{0, 1\}^{6 \times 61}$, where $(A)_{i,c} = 1$ if $c \in \mathcal{C}_i$ and 0 otherwise. Then $a_g = A c_g$.

Each row of A has disjoint support and entries in $\{0, 1\}$, so

$$(AA^\top)_{i,i} = \sum_c (A_{i,c})^2 = |\mathcal{C}_i|, \quad (AA^\top)_{i,j} = 0 \quad (i \neq j).$$

Thus $AA^\top = \text{diag}(|\mathcal{C}_1|, \dots, |\mathcal{C}_6|)$, and the operator norm of A satisfies

$$\|A\|_{2 \rightarrow 2}^2 = \lambda_{\max}(AA^\top) = \max_i |\mathcal{C}_i|.$$

In practice each $|\mathcal{C}_i|$ is finite and modest, and since $\sum_c c_g(c) = 1$ with all components nonnegative, we have $\|c_g\|_2 \leq \sqrt{\sum_c c_g(c)^2} \leq \sqrt{\sum_c c_g(c)} = 1$. Hence

$$\|a_g\|_2 = \|Ac_g\|_2 \leq \|A\|_{2 \rightarrow 2} \|c_g\|_2 \leq \sqrt{\max_i |\mathcal{C}_i|}.$$

Because $\max_i |\mathcal{C}_i|$ is a constant (independent of g), we can absorb this into the global constant of the Track A operator-norm budget. For scaling convenience, one can simply renormalize A to ensure $\|A\|_{2 \rightarrow 2} \leq 1$, which yields $\|a_g\|_2 \leq \|c_g\|_2 \leq 1$. $\square$

B.2 Composition with 6-Channel WHT

Recall $h_g = H_6 a_g$, where H_6 is orthogonal.

Proposition B.3 (Overall spectral norm bound). *For each gene g ,*

$$\|h_g\|_2 \leq \|c_g\|_2 \leq 1.$$

Proof. Using the previous bounds and orthogonality of H_6 ,

$$\|h_g\|_2 = \|H_6 a_g\|_2 \leq \|H_6\|_{2 \rightarrow 2} \|a_g\|_2 = \|a_g\|_2,$$

since $\|H_6\|_{2 \rightarrow 2} = 1$ for an orthogonal matrix. Taking $\|a_g\|_2 \leq \|c_g\|_2 \leq 1$ from the previous proposition yields the claim. $\square$

C GC3-Controlled Null Ensemble and Δh

C.1 Null Ensemble Construction

For each gene g , the GC3-controlled null ensemble consists of random codon-usage realizations preserving the amino-acid sequence and GC3 content.

Definition C.1 (Null ensemble). *Fix a number of replicates $R \in \mathbb{N}$. For gene g , let $\{c_{g,r}^{\text{null}}\}_{r=1}^R$ be an i.i.d. sample of codon-usage vectors constructed by GC3-preserving codon reshuffling per position. Define the corresponding aggregated spectra*

$$h_{g,r}^{\text{null}} = H_6 A c_{g,r}^{\text{null}}, \quad r = 1, \dots, R.$$

Definition C.2 (Mean null spectrum and deviation). *The mean null spectrum is*

$$\bar{h}_g^{\text{null}} = \frac{1}{R} \sum_{r=1}^R h_{g,r}^{\text{null}}.$$

Define the deviation vector

$$\Delta h_g = h_g - \bar{h}_g^{\text{null}} \in \mathbb{R}^6,$$

and its scalar magnitude

$$\delta_g = \|\Delta h_g\|_2.$$

C.2 Norm Bounds for Δh

Proposition C.3 (Deviation norm bound). *For each gene g ,*

$$\|\Delta h_g\|_2 \leq \|h_g\|_2 + \|\bar{h}_g^{\text{null}}\|_2 \leq 2.$$

Proof. By the triangle inequality,

$$\|\Delta h_g\|_2 = \|h_g - \bar{h}_g^{\text{null}}\|_2 \leq \|h_g\|_2 + \|\bar{h}_g^{\text{null}}\|_2.$$

Each null spectrum $h_{g,r}^{\text{null}}$ has the same norm bound as h_g because GC3-preserving reshuffles are still normalized codon-usage distributions; hence $\|h_{g,r}^{\text{null}}\|_2 \leq 1$ using the previous section's bound. Therefore,

$$\|\bar{h}_g^{\text{null}}\|_2 = \left\| \frac{1}{R} \sum_{r=1}^R h_{g,r}^{\text{null}} \right\|_2 \leq \frac{1}{R} \sum_{r=1}^R \|h_{g,r}^{\text{null}}\|_2 \leq 1.$$

Combining with $\|h_g\|_2 \leq 1$ yields $\|\Delta h_g\|_2 \leq 2$. $\square$

Corollary C.4 (Bound on scalar δ_g). *For each gene g , the scalar deviation satisfies*

$$0 \leq \delta_g \leq 2.$$

Proof. By definition $\delta_g = \|\Delta h_g\|_2 \geq 0$. The upper bound follows directly from the previous proposition. $\square$

Remark C.5. *In practice, because both h_g and $\bar{h}_g^{\text{null}}$ are weighted combinations of channels constructed from a probability vector, typical norms will be substantially below 1; the bound above is worst-case and sufficient for numerical stability and feature scaling guarantees in the Track A logistic models.*

D Logistic Regression and Feature Scaling

D.1 Feature Space and Scaling

Let $y_g \in \{0, 1\}$ be the binary label for gene g , and consider two feature sets:

$$X_g^{\text{GC3}} = (\text{GC3}_g) \in \mathbb{R}, \quad X_g^{\text{GC3}+\delta} = (\text{GC3}_g, \delta_g) \in \mathbb{R}^2,$$

or the vector form

$$X_g^{\text{GC3}+\Delta h} = (\text{GC3}_g, \Delta h_{g,1}, \dots, \Delta h_{g,6}) \in \mathbb{R}^7.$$

From the definition of GC3, we have $0 \leq \text{GC3}_g \leq 1$ for all genes. Combined with $0 \leq \delta_g \leq 2$, all coordinates of Track A feature vectors lie in a compact interval. This uniform boundedness is useful for controlling the Lipschitz constants of logistic losses and for regularization tuning.

Proposition D.1 (Uniform boundedness of features). *For all genes g ,*

$$\|X_g^{\text{GC3}}\|_2 \leq 1, \quad \|X_g^{\text{GC3}+\delta}\|_2 \leq \sqrt{1^2 + 2^2} = \sqrt{5},$$

and similarly $\|X_g^{\text{GC3}+\Delta h}\|_2 \leq \sqrt{1 + 6 \cdot 2^2} = \sqrt{25} = 5$.

Proof. Immediate from the coordinate bounds: $\text{GC3}_g \in [0, 1]$, each $\Delta h_{g,i}$ component has magnitude at most $\|\Delta h_g\|_2 \leq 2$, and $\delta_g \leq 2$. $\square$

D.2 Logistic Loss Lipschitz Bound

Consider the logistic regression model with parameter vector $\theta \in \mathbb{R}^d$ and features $X_g \in \mathbb{R}^d$,

$$p_\theta(y_g = 1 \mid X_g) = \sigma(\theta^\top X_g), \quad \sigma(z) = \frac{1}{1 + e^{-z}}.$$

Define the negative log-likelihood for a single sample:

$$\ell(\theta; X_g, y_g) = -y_g \log \sigma(\theta^\top X_g) - (1 - y_g) \log(1 - \sigma(\theta^\top X_g)).$$

Proposition D.2 (Gradient Lipschitz constant). *Let X_g be any of the Track A feature vectors described above, with $\|X_g\|_2 \leq B$ for all g . Then the gradient of ℓ with respect to θ satisfies*

$$\|\nabla_\theta \ell(\theta; X_g, y_g)\|_2 \leq B.$$

Proof. The gradient is

$$\nabla_\theta \ell(\theta; X_g, y_g) = (\sigma(\theta^\top X_g) - y_g) X_g.$$

Since $\sigma(\theta^\top X_g) \in (0, 1)$ and $y_g \in \{0, 1\}$, we have $|\sigma(\theta^\top X_g) - y_g| \leq 1$. Therefore

$$\|\nabla_\theta \ell(\theta; X_g, y_g)\|_2 = |\sigma(\theta^\top X_g) - y_g| \|X_g\|_2 \leq \|X_g\|_2 \leq B.$$

□

Corollary D.3 (Global Lipschitz bound). *Over the Track A feature sets, the logistic loss gradient is uniformly bounded with $B = 5$ in the worst case (GC3+ Δh vector model), which yields global step-size conditions for first-order optimization methods.*

E Contraction Arguments for Feedback Variants

While Track A in its simplest form does not require dynamic feedback, the same mathematical framework used in Track B can be restricted to the purely classical Track A setting, should adaptive weighting or iterative re-labeling be introduced.

E.1 Static vs. Dynamic Interpretation

In a dynamic interpretation, one might define a Track A state $S_t^{(A)} = (c_{g,t}, h_{g,t}, \Delta h_{g,t})$ evolving under iterative refinement (e.g., repeated null-ensemble updates or re-weighting of channels based on current classifier residuals).

Definition E.1 (Track A feedback map (hypothetical)). *Let $Z_t^{(A)}$ denote a vector of Track A state descriptors (e.g., averaged spectra, residuals, etc.). A feedback map is a function*

$$f_A : \mathbb{R}^d \times \mathbb{R}^k \rightarrow \mathbb{R}^d, \quad Z_{t+1}^{(A)} = f_A(Z_t^{(A)}, \Xi_t),$$

where Ξ_t encodes observed statistics (e.g., accuracy, cross-entropy, etc.) at iteration t .

Theorem E.2 (Contraction condition for Track A feedback). *Suppose there exists $L < 1$ such that*

$$\|f_A(Z, \Xi) - f_A(Z', \Xi)\|_2 \leq L \|Z - Z'\|_2$$

for all Z, Z' in the state space and for all admissible Ξ . Then:

1. There exists a unique fixed point $Z_*^{(A)}$.
2. For any initialization $Z_0^{(A)}$, the iterates converge geometrically:

$$\left\| Z_t^{(A)} - Z_*^{(A)} \right\|_2 \leq L^t \left\| Z_0^{(A)} - Z_*^{(A)} \right\|_2.$$

Proof. This is a direct application of the Banach Fixed-Point Theorem on the complete metric space $(\mathbb{R}^d, \|\cdot\|_2)$. $\square$

Although Track A as specified is a single-pass pipeline, this result is included to document the operator-norm and contraction logic underlying any future adaptive extensions of the protocol.

F Summary of Operator Norm Bounds

For convenience, we collect the key norm bounds proved above.

- **Hadamard/WHT norm preservation.** For normalized Hadamard matrices $\hat{H}_n$ and the 6-channel orthogonal matrix H_6 ,

$$\left\| \hat{H}_n x \right\|_2 = \|x\|_2, \quad \|H_6 a\|_2 = \|a\|_2.$$

- **Codon aggregation bound.** With appropriate normalization of the aggregation matrix A ,

$$\|a_g\|_2 \leq \|c_g\|_2 \leq 1.$$

- **6-channel spectrum bound.** For each gene,

$$\|h_g\|_2 \leq \|c_g\|_2 \leq 1.$$

- **Null ensemble deviation.** For each gene,

$$0 \leq \delta_g = \left\| h_g - \bar{h}_g^{\text{null}} \right\|_2 \leq 2.$$

- **Feature vector bounds.** For GC3, GC3+ δ , and GC3+ Δh feature vectors,

$$\|X_g^{\text{GC3}}\|_2 \leq 1, \quad \|X_g^{\text{GC3}+\delta}\|_2 \leq \sqrt{5}, \quad \|X_g^{\text{GC3}+\Delta h}\|_2 \leq 5.$$

- **Logistic loss gradient.** The gradient of the per-sample logistic loss satisfies

$$\|\nabla_{\theta} \ell(\theta; X_g, y_g)\|_2 \leq B,$$

with $B \in \{1, \sqrt{5}, 5\}$ depending on the feature set.

These results provide the mathematical backbone for Track A’s numerical stability, scaling behavior, and potential extension to adaptive or iterative schemes.

References (for background only)

- Standard descriptions of Hadamard/Walsh–Hadamard transforms and their orthonormality properties.[?, ?, ?]
- Overviews of codon usage analysis and GC3-driven models, highlighting the central role of GC3 in codon usage variation.[?, ?, ?]

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